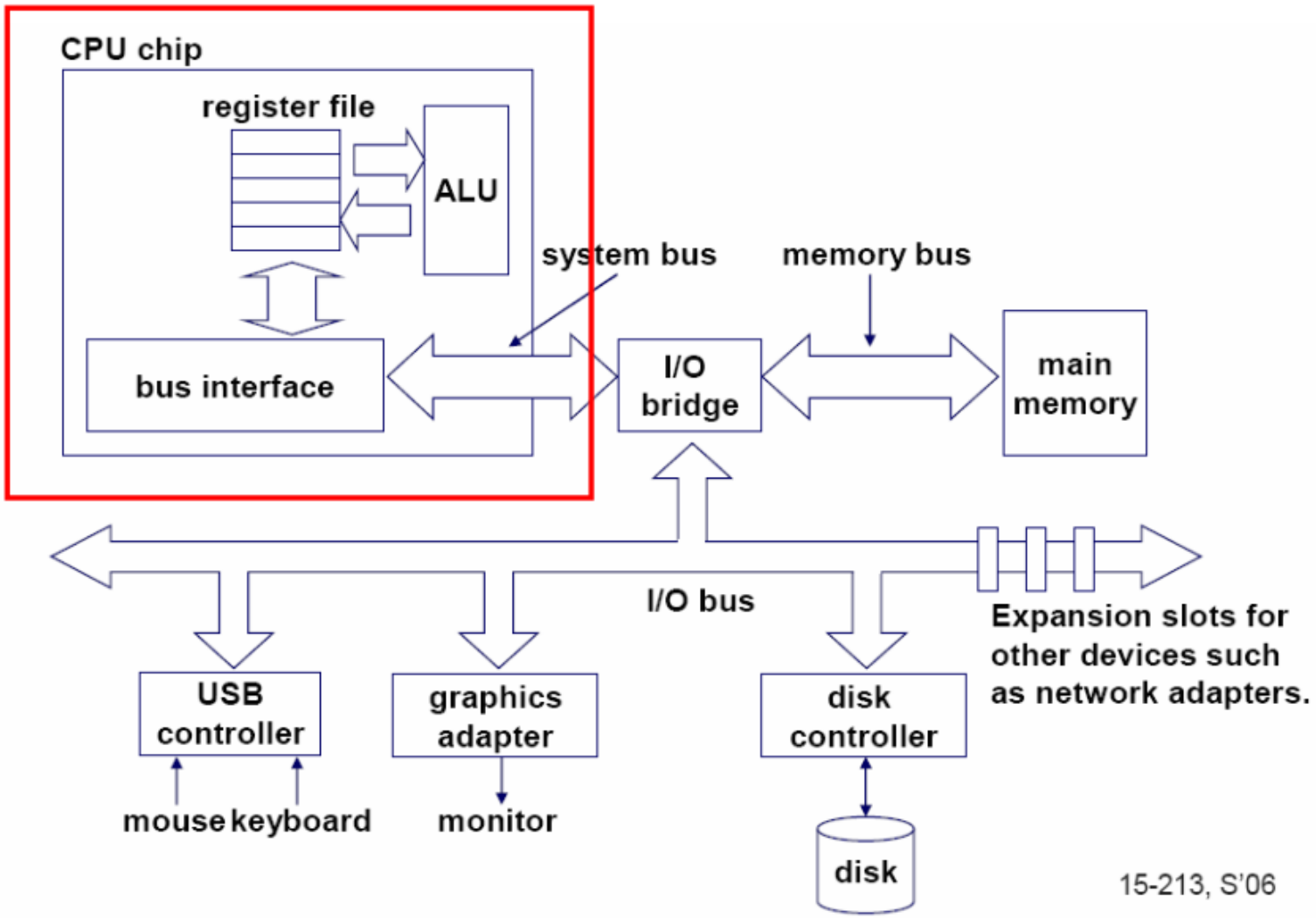
MULTICORE ARCHITECTURES

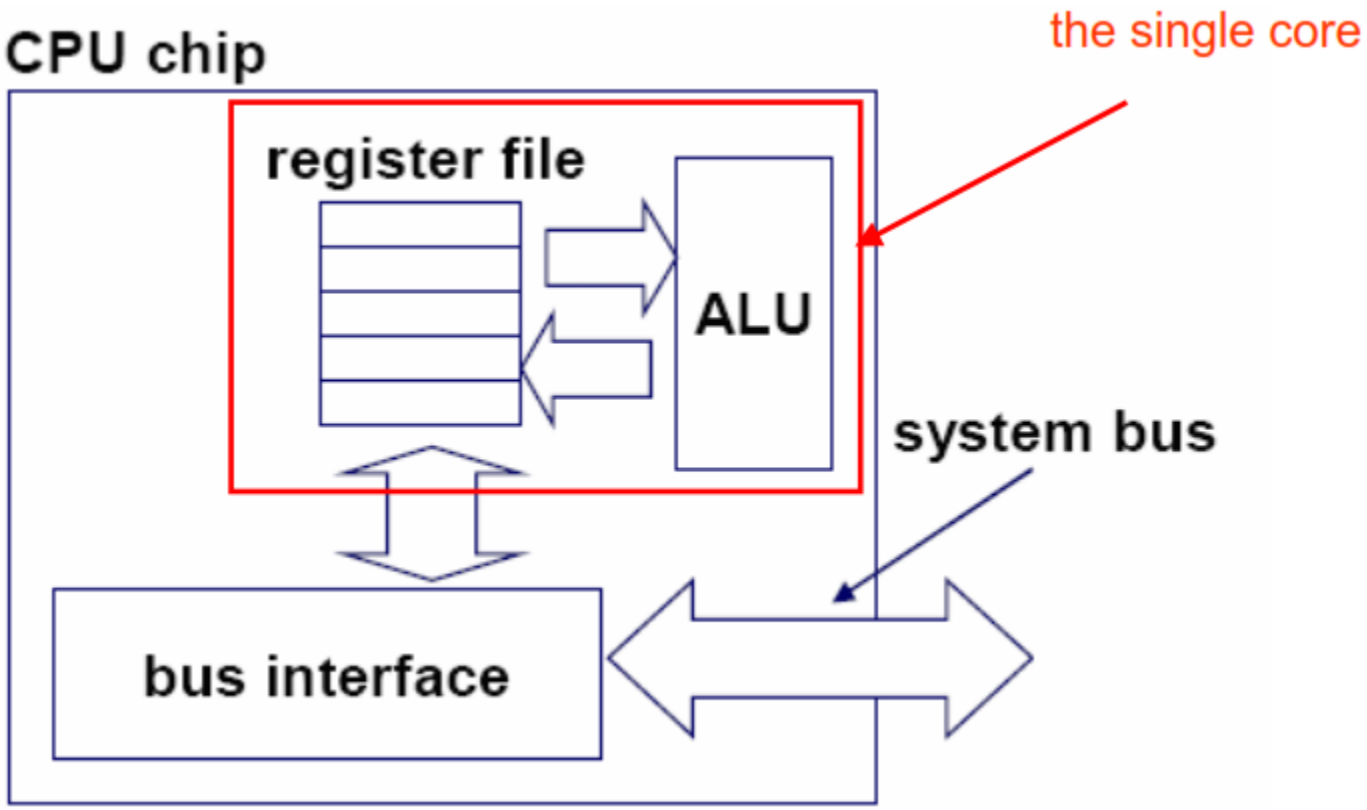
Dr Noor Mahammad Sk

Single-core computer

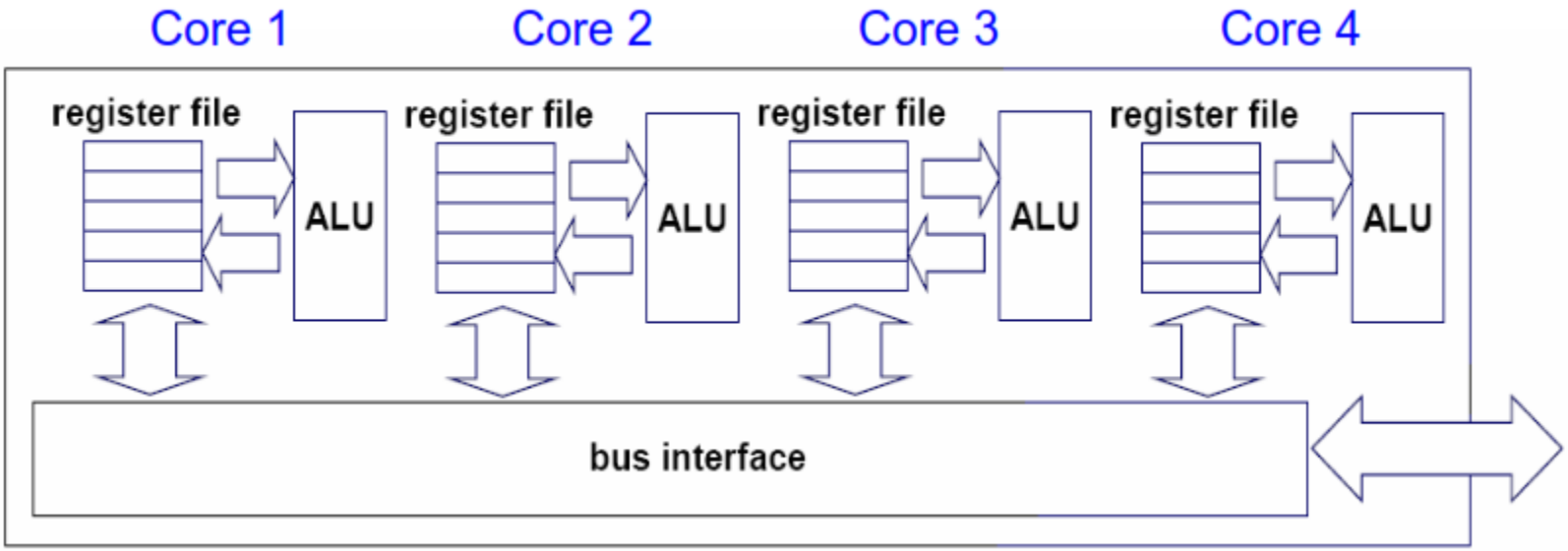


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Single Core CPU Chip

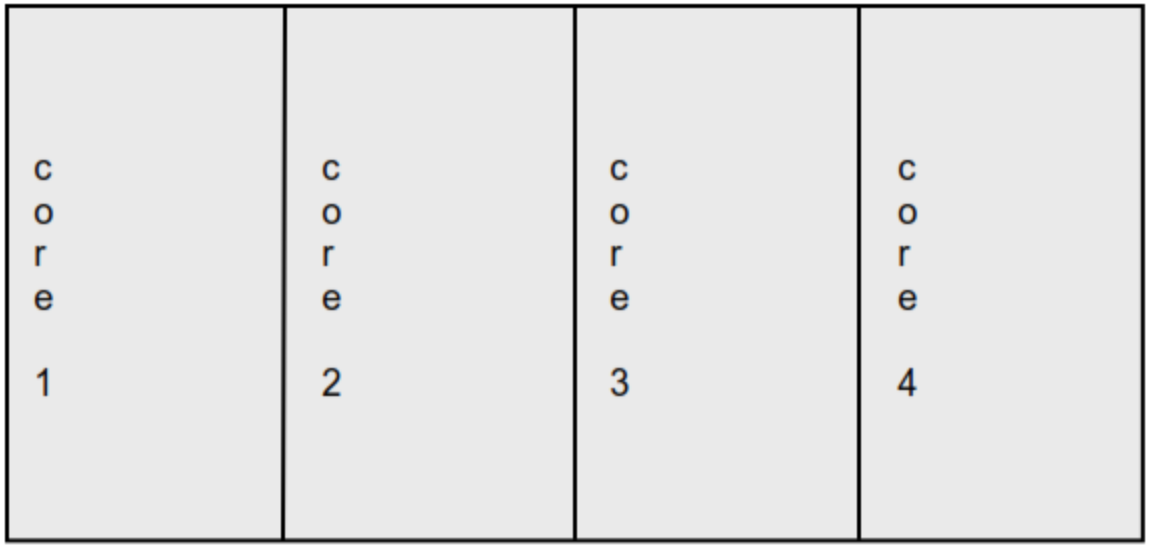


Multi-core architectures

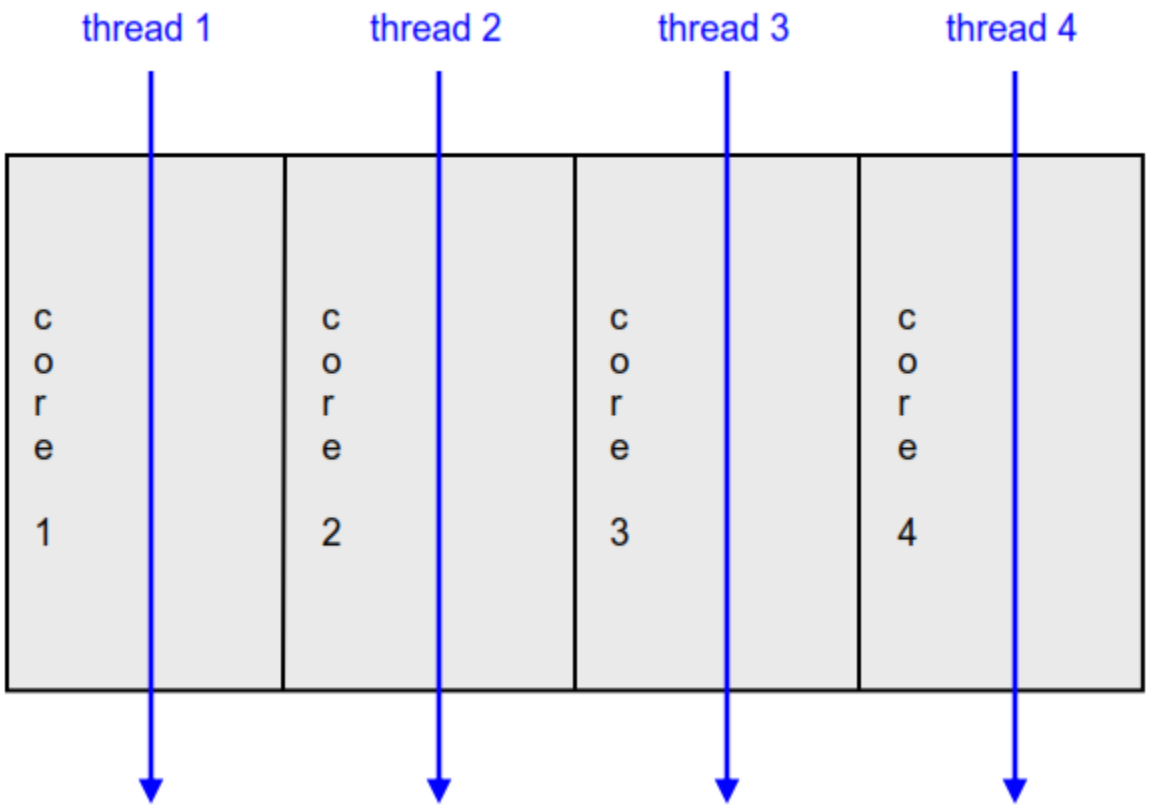


Multi-core CPU chip

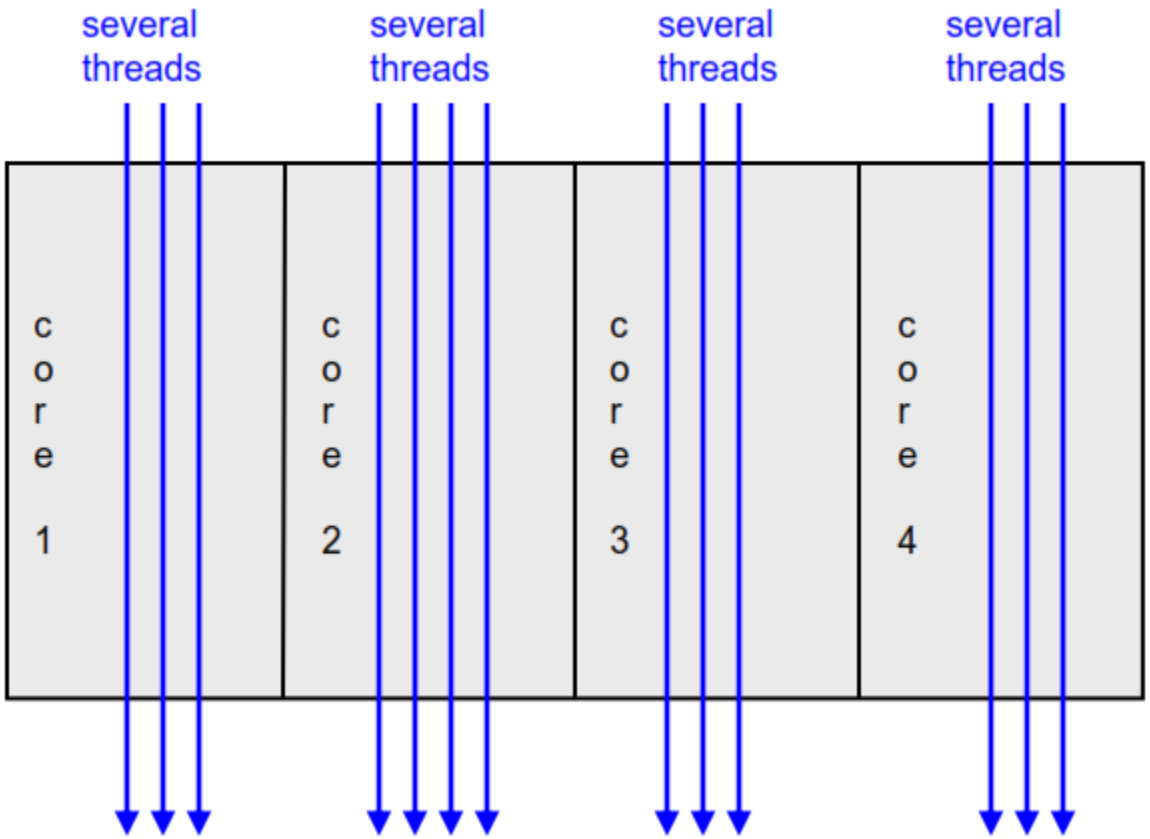
- The cores fit on a single processor socket
- Also called CMP (Chip Multi-Processor)



The cores run in parallel



Within each core, threads are time-sliced (just like on a uniprocessor)



Interaction with OS

- OS perceives each core as a separate processor
- OS scheduler maps threads/processes to different cores
- Most major OS support multi-core today

Why multi-core?

- Difficult to make single-core clock frequencies even higher
- Limitation of ILP
- Many new applications are multithreaded
- General trend in computer architecture (shift towards more parallelism)

Instruction-level parallelism

- Parallelism at the machine-instruction level
- The processor can re-order, pipeline instructions, split them into microinstructions, do aggressive branch prediction, etc.

Thread-level parallelism (TLP)

- This is parallelism on a more coarser scale
- Server can serve each client in a separate thread (Web server, database server)
- A computer game can do AI, graphics, and physics in three separate threads
- Single-core superscalar processors cannot fully exploit TLP
- Multi-core architectures are the next step in processor evolution: explicitly exploiting TLP

General context: Multiprocessors

- Multiprocessor is any computer with several processors
- SIMD
  - Single instruction, multiple data
  - Modern graphics cards
- MIMD
  - Multiple instructions, multiple data

Multiprocessor memory types

- Shared memory:
- In this model, there is one (large) common shared memory for all processors
- Distributed memory:
- In this model, each processor has its own (small) local memory, and its content is not replicated anywhere else

Multi-core processor is a special kind of a multiprocessor: All processors are on the same chip

- Multi-core processors are MIMD:
- Different cores execute different threads (Multiple Instructions), operating on different parts of memory (Multiple Data).
- Multi-core is a shared memory multiprocessor:
- All cores share the same memory

What applications benefit from multi-core?

- Database servers
- Web servers (Web commerce)
- Compilers
- Multimedia applications
- Scientific applications,
- CAD/CAM
- In general, applications with *Thread-level parallelism*

(as opposed to instruction-level parallelism)

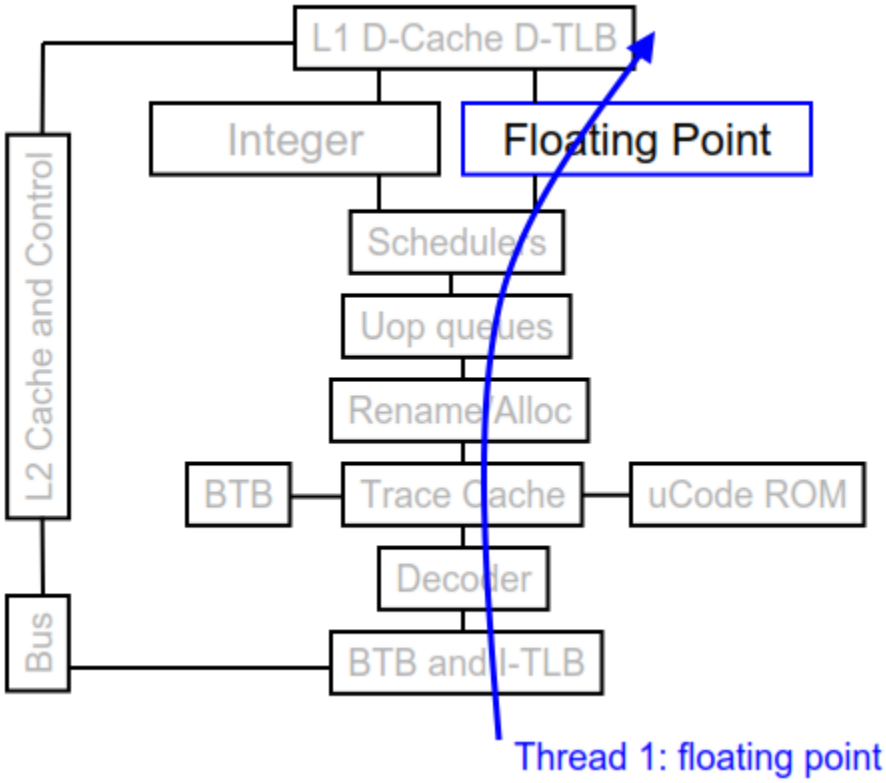
More examples

- Editing a photo while recording a TV show through a digital video recorder
- Downloading software while running an anti-virus program
- “Anything that can be threaded today will map efficiently to multi-core”
- BUT: some applications difficult to parallelize

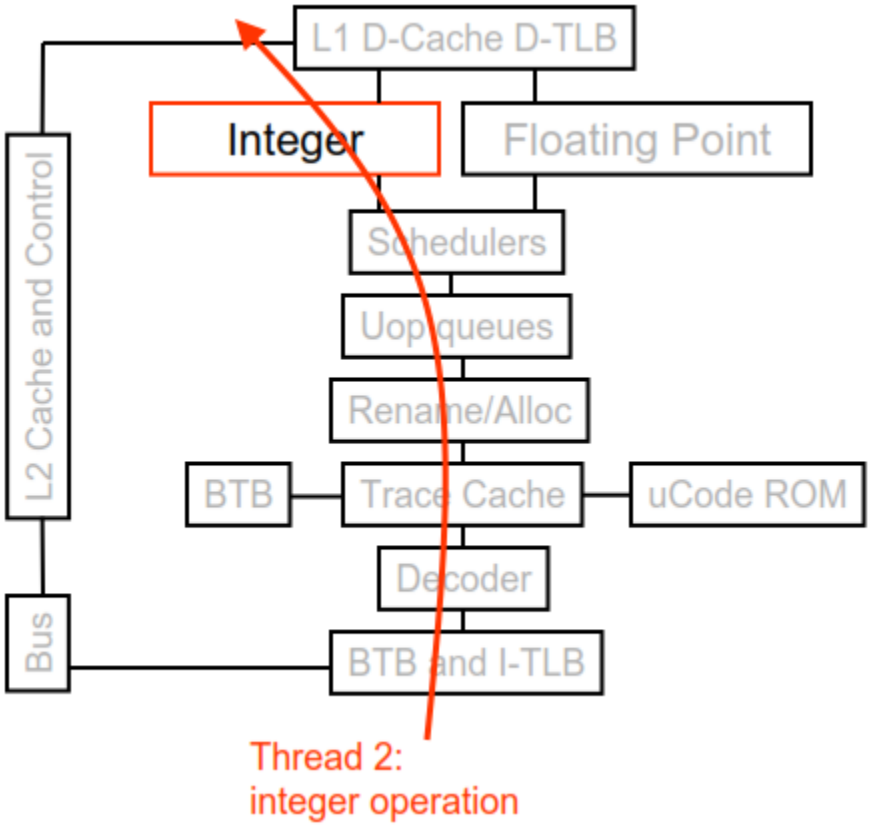
Simultaneous multithreading (SMT)

- Permits multiple independent threads to execute SIMULTANEOUSLY on the SAME core
- Weaving together multiple “threads” on the same core
- Example: if one thread is waiting for a floating point operation to complete, another thread can use the integer units

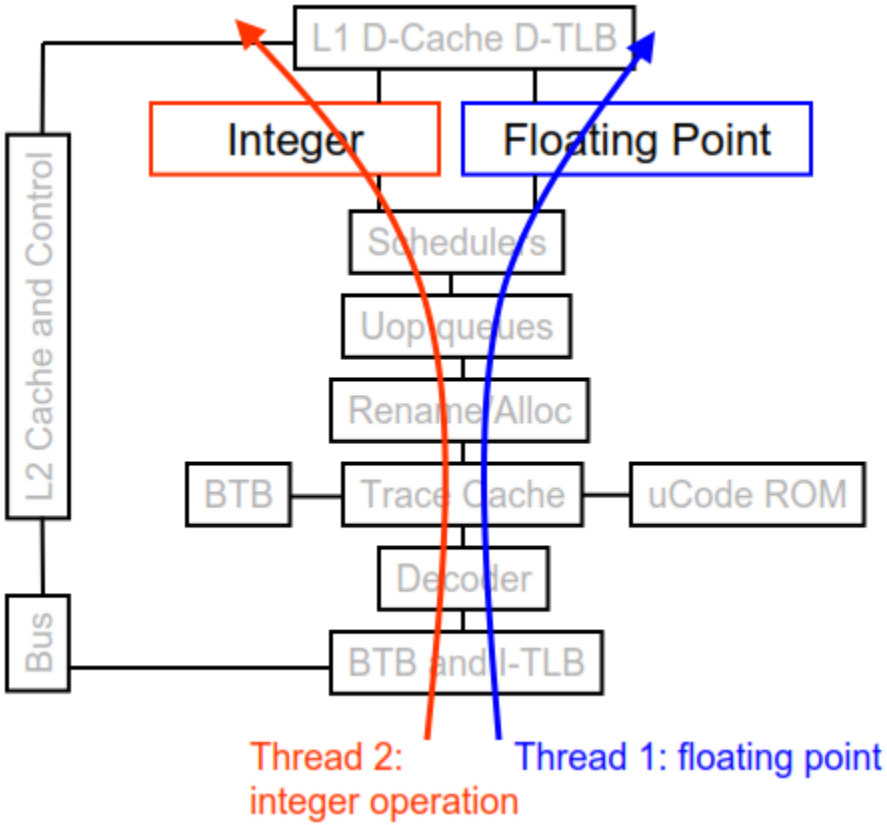
Without SMT, only a single thread can run at any given time



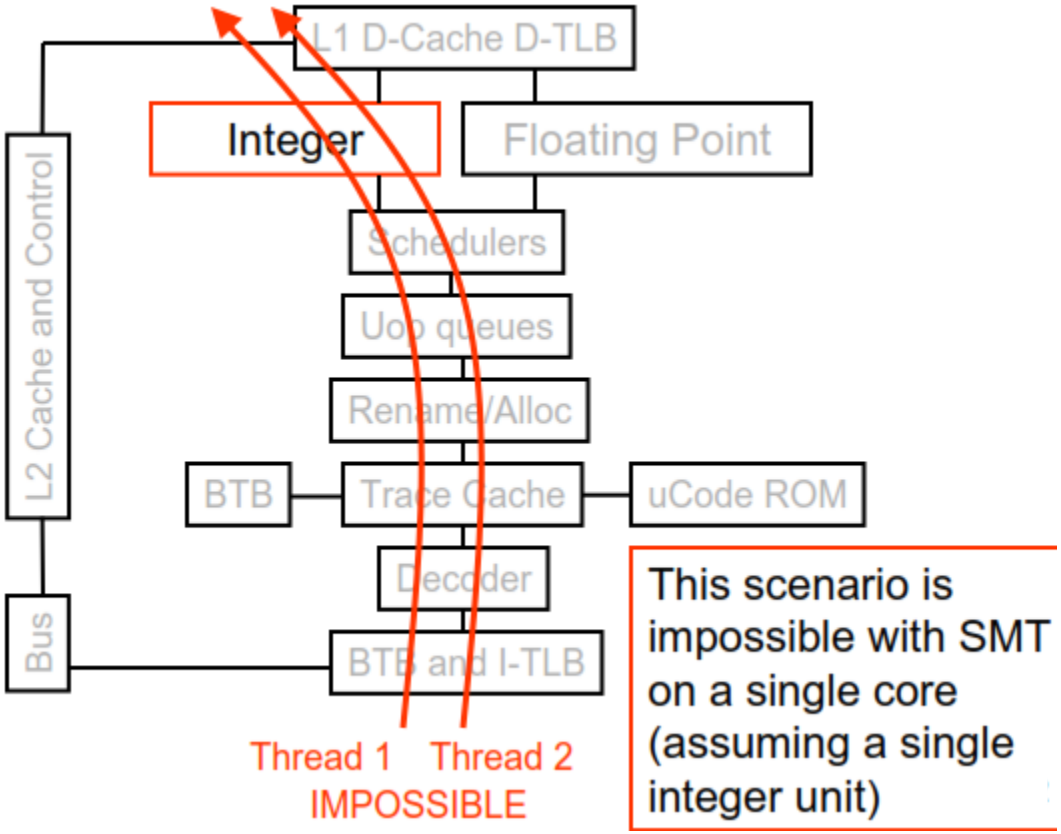
Without SMT, only a single thread can run at any given time



SMT processor: both threads can run concurrently



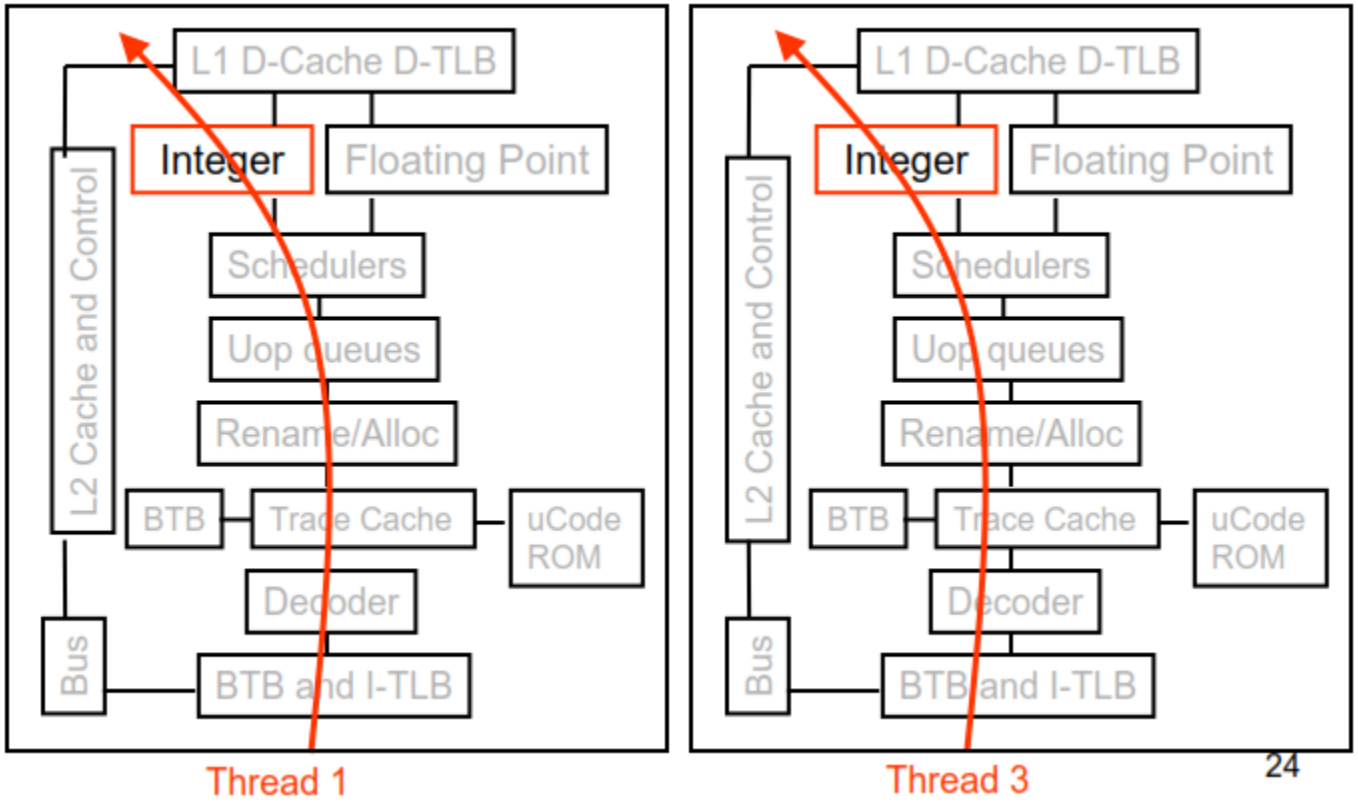
But: Can't simultaneously use the same functional unit



SMT not a “true” parallel processor

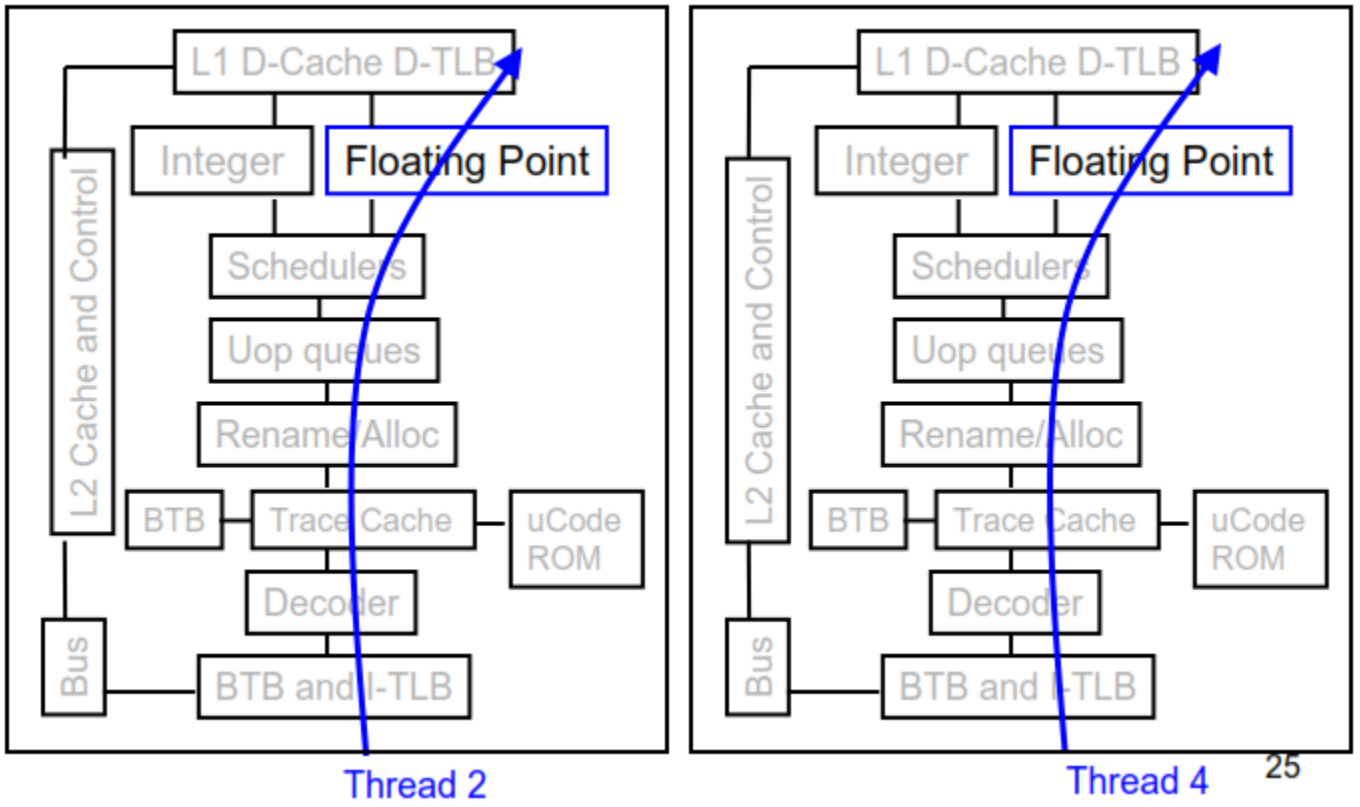
- Enables better threading (e.g. up to 30%)
- OS and applications perceive each simultaneous thread as a separate “virtual processor”
- The chip has only a single copy of each resource
- Compare to multi-core: each core has its own copy of resources

Multi-core:
threads can run on separate cores



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Multi-core:
threads can run on separate cores



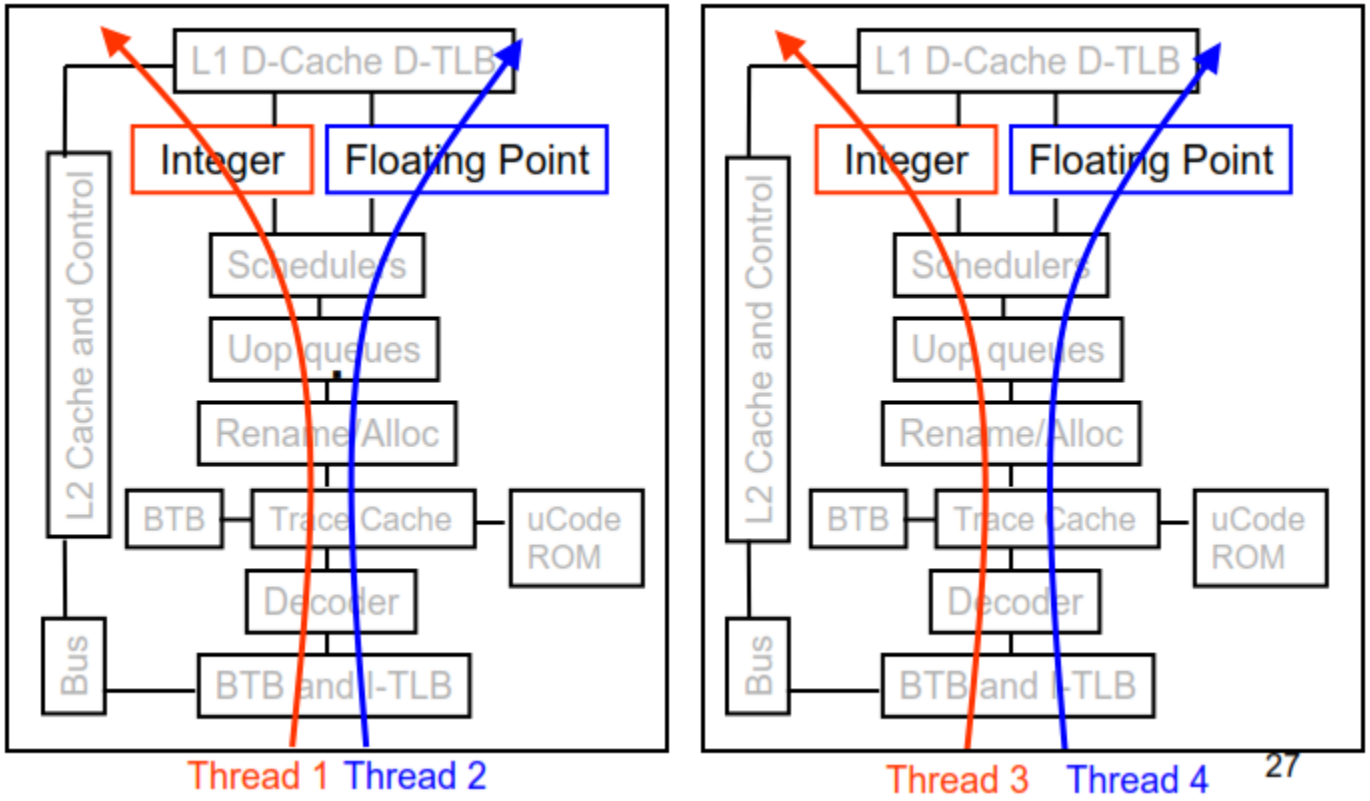
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Combining Multi-core and SMT

- Cores can be SMT-enabled (or not)
- The different combinations:
  - Single-core, non-SMT: standard uniprocessor
  - Single-core, with SMT
  - Multi-core, non-SMT
  - Multi-core, with SMT: our fish machines
- The number of SMT threads:

  2, 4, or sometimes 8 simultaneous threads
- Intel calls them “hyper-threads”

SMT Dual-core: all four threads can run concurrently



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Comparison: multi-core vs SMT

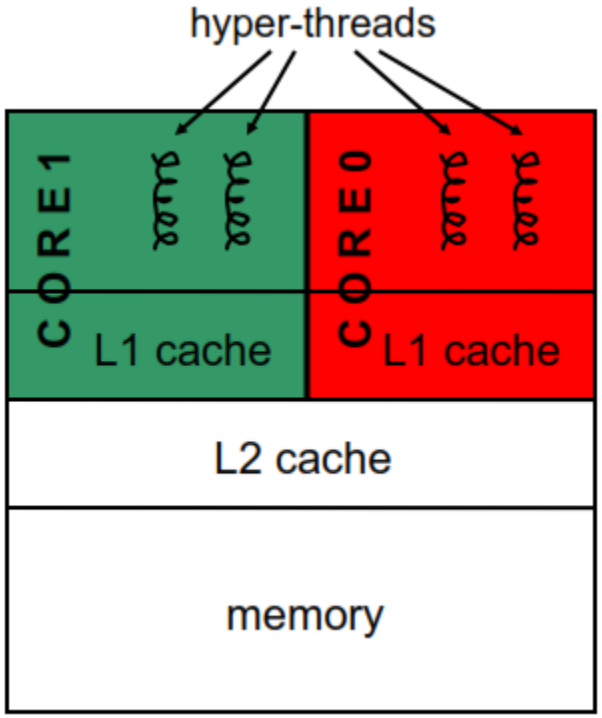
- Multi-core:
  - Since there are several cores, each is smaller and not as powerful (but also easier to design and manufacture)
  - However, great with thread-level parallelism
- SMT
  - Can have one large and fast superscalar core
  - Great performance on a single thread
  - Mostly still only exploits instruction-level parallelism

The memory hierarchy

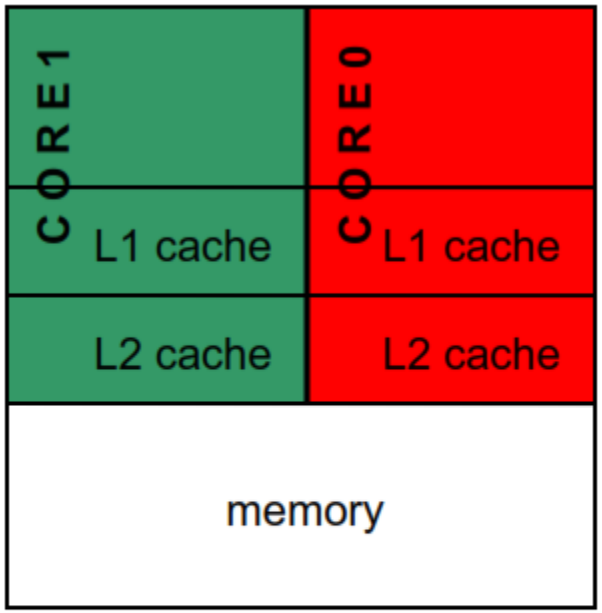
- If simultaneous multithreading only:
  - all caches shared
- Multi-core chips:
  - L1 caches private
  - L2 caches private in some architectures and shared in others
- Memory is always shared

Memory Hierarchy

- Dual-core Intel Xeon processors
- Each core is hyper-threaded
- Private L1 caches
- Shared L2 caches

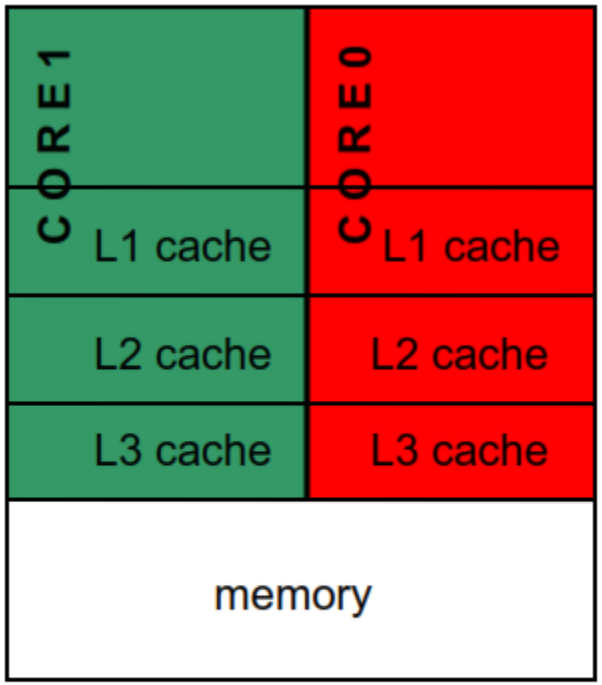


Designs with private L2 caches



Both L1 and L2 are private

Examples: AMD Opteron, AMD Athlon, Intel Pentium D



A design with L3 caches

Example: Intel Itanium 2

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